

Milestone 1 report

Smart MCQ Solver Challenge

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1. Scope

Milestone 1 deliverables:

- Reproducible wrapper-stripping regex with unit tests (`src/normalize.py`, `tests/test_normalize.py`).
- EDA notebook documenting the closed-question-bank finding (`notebooks/m1_eda.ipynb`).
- Workstream A v1 lookup pipeline with PH-A placeholder for the 45 paraphrase rows (`notebooks/m1_workstream_a_v1.ipynb`).
- Kaggle inference notebook port with diagnostic cells (`notebooks/kaggle_inference_v1.ipynb`).
- First competition submission to establish the LB floor.
- Revised dataset findings reflecting the actual LB ceiling (`reports/dataset_findings_revised.md`).

Workstream B (trained models) was not in M1 scope.

2. Dataset

`train.csv` contains 2,000 rows and 8 columns: `id`, `prompt`, options A through E, and `answer`. No null values appear in any option column. Prompt length ranges from 3 to 51 whitespace-tokens (mean 18.1, median 17). The five option columns have nearly identical length distributions; per-column mean falls between 26.0 and 26.5 tokens. `test.csv` shares the same schema minus the `answer` column.

Structurally, `train` covers 252 unique normalised stems at roughly 8x replication per stem with re-randomised wrappers. `test` covers 228 unique normalised stems, all of which appear in `train`. 455 of 500 test rows (91%) match a `train` row on both stem and option text. The remaining 45 rows (9%) share a stem with `train` but have rephrased option text. All 222 multi-occurrence `train` stems carry one consistent answer label across duplicates. The wrapper inventory is 6 prefixes and 4 suffixes; iterative stripping is required for compound cases. After full stripping, every test stem matches a `train` stem.

The structural facts reproduce both locally and on the Kaggle runtime. Diagnostic cells in `notebooks/kaggle_inference_v1.ipynb` confirm the 2,000/500 shapes, 252/228 unique stems, 500/500 stem coverage, and 455 identity-match rows.

3. Answer distribution

Gold-label counts across the 2,000 training rows:

Label	Count	Share
A	369	18.45%
B	490	24.50%
C	459	22.95%
D	358	17.90%

Label	Count	Share
E	324	16.20%

Most frequent (B = 490) plus least frequent (E = 324) sums to 814. The distribution skews toward B and C, so a random-guess prior of 0.2 is not a fair upper bound for trivial baselines.

4. EDA: vocabulary and stop-word filtering

Procedure: lowercase each prompt, strip every character in `string.punctuation`, split on whitespace. The union of tokens across all 2,000 cleaned prompts yields 859 unique words. The small size reflects templated prompt phrasing; many begin with stems such as "What is..." or "Pick the best possible answer...".

The row-1 prompt contains 22 tokens after cleaning. After dropping `sklearn.feature_extraction.text.ENGLISH_STOP_WORDS`, 13 tokens remain.

5. TF-IDF vocabulary (Q4)

A default `TfidfVectorizer(stop_words='english')` was fit on "all the combined text of the prompts and options in `train.csv`." That phrasing admits two readings, both implemented to confirm identical vocabulary size.

Reading A: each prompt and each option is a separate document (2,000 + 2,000 x 5 = 12,000 documents). Reading B: each row contributes one document built by concatenating its prompt with all five options (2,000 documents). Both readings produce 2,762 feature columns. IDF weights differ between the two fits because document counts differ, so any cosine score computed downstream depends on which reading is used.

6. Cosine similarity, row 1, prompt vs option A (Q5)

`cosine_similarity(prompt, A)` for row 1:

Reading	Value
A (12,000 documents)	0.2328
B (2,000 documents)	0.2720

Reported submission: 0.2720 (reading B). Reading B is canonical because "combined text of the prompts and options" most naturally describes per-row concatenation rather than a flat pool of texts.

7. Top-1 cosine accuracy (Q6)

For each row, the option with the highest cosine similarity to the prompt is selected and compared against the gold answer.

Reading	Top-1 accuracy
A	13.70%
B	13.55%

Both figures fall below the 20% uniform-random rate. 342 rows (17.1%) produce a zero-cosine vector against every option after stop-word removal, because the cleaned prompt shares no surviving tokens with any option. In those rows the prediction falls back on tie-breaking by option-letter order. TF-IDF on short prompts

emphasises rare prompt tokens, and the correct option rarely repeats those tokens literally. A "longest option" baseline reaches 34.95% top-1 accuracy on the same data, confirming that TF-IDF cosine is anti-predictive here.

8. MAP@3 worked examples (Q7, Q8)

MAP@3 with a single relevant label and a ranked list of three predictions: the contribution is $1/\text{rank}$ if the gold label appears in the top three, otherwise zero.

Gold label	Predictions	Rank	Score
C	[C, A, B]	1	$1/1 = 1.0$
B	[D, B, E]	2	$1/2 = 0.5$

9. Majority-class baseline (Q9)

Sort answer counts to fix a single static ranking for every row: [B, C, A]. Apply this ranking to all 2,000 rows and average the per-row MAP@3.

Closed-form: $(c_B \times 1/1 + c_C \times 1/2 + c_A \times 1/3) / N = (490 + 229.5 + 123) / 2,000 = 842.5 / 2,000 = 0.4213$. The implemented loop returns 0.4212, consistent with floating-point rounding.

10. TF-IDF pipeline MAP@3 (Q10)

For each row, all five options are scored by cosine similarity against the prompt, ranked descending, and the top three submitted. Tie-breaking uses a stable argsort, resolving equal similarities by original column order (A, B, C, D, E).

Reading	MAP@3
A	0.3119
B	0.2963

Reported submission: 0.2963 (reading B). The TF-IDF pipeline underperforms the majority-class baseline (0.4212) for the same reason flagged in section 7: lexical overlap between prompt and option is a poor proxy for correctness, and the prompts are too short for TF-IDF to recover meaningful weights.

11. Workstream A v1 pipeline

Deterministic. No training.

Step 1. Apply `normalize_stem` (iterative regex strip) to both train and test prompts.

Step 2. Build two structures from train: `train_lookup` maps `(stem, frozenset(options))` to the correct option text, handling the 455 identity-match rows; `stem_to_answer` maps stem to answer label and serves as the PH-A fallback for the 45 paraphrase rows.

Step 3. For each test row: if its signature `(stem, frozenset(options))` appears in `train_lookup`, locate the correct option text inside the test row's options and predict its label. Otherwise (paraphrase residual), predict the train stem's answer label directly (PH-A).

Step 4. Top-3 construction: predicted label first, then two filler labels from the global label prior (B, C, A, D, E in that order, skipping the predicted label).

Step 5. Self-check: applying the same lookup to train recovers all 2,000 train answer labels (100%).

PH-A justification: on the 45 paraphrase rows, fuzzy text similarity between each test option and the train correct option text agrees with the train stem's answer label in 43 / 45 cases (95.56%), above the 0.80 threshold to prefer PH-A over PH-B (global prior baseline).

12. M1 submission result

Metric	Value
Public LB score	0.74147
Rank at submission	56 / 112 registered
Submission file	submission.csv written to /kaggle/working/
Diagnostic checks	Shapes, unique-stem counts, identity-match count, and top-1 distribution all match local analysis.

13. Strategic finding: corrected lookup ceiling

The preliminary findings document predicted that naive lookup with wrapper stripping would score 0.90+ MAP@3 with a competitive ceiling near 0.97-0.99. The LB contradicts this:

Rank	Score
1 / 2 / 3 (tied)	0.76475
4	0.76226
5	0.75935
Ours (56)	0.74147

Three independent submitters arriving at the same score (0.76475) indicates this is the deterministic lookup ceiling. Our submission sits at approximately 97% of that ceiling.

Most plausible explanation: test answer keys were generated independently of train. For roughly 23-25% of test rows the scorer's correct answer disagrees with the train label even when stem and options match byte-exactly. Pure lookup cannot recover these rows. Full revision in reports/dataset_findings_revised.md.

14. Pivot for M2

Workstream A is near-optimal at 0.74. Further lookup polish (TTA option permutation, fuzzy paraphrase matching, smarter filler) has bounded upside (approximately 0.02 LB) and is deferred to a later opportunistic pass.

Workstream B is promoted from a course-grade obligation to the primary LB lever. A trained model that reasons over each question independently can in principle answer correctly on the roughly 25% of rows where the train label is wrong from the scorer's perspective.

M2 plan:

- First Workstream B model: DeBERTa-v3-base with per-option encoding and shared backbone, softmax over 5 option scores.

- Stem-grouped GroupKFold validation (no train/val stem overlap); val MAP@3 reflects pure inferential ability.
- W&B; logging: loss curves, val MAP@3 per fold, learning-rate schedule, hyperparameters, gradient norms.
- The originally planned M2 paraphrase fuzzy matcher is demoted to opportunistic polish, executed only if individual Workstream B models converge above 0.74 and an ensemble could plausibly lift further.

15. Open items carried into M2

- Set up the M2 training notebook scaffold (Kaggle GPU or Colab).
- Confirm DeBERTa-v3-base weights source (HuggingFace).
- Decide tokenisation strategy under the 512-token window given max option length 662 chars; per-option encoding handles this naturally.
- Establish W&B; run naming convention and project structure for Workstream B.
- Stem-grouped GroupKFold split: 5 folds, deterministic seed, fold assignment cached so subsequent models train on identical splits.

None of the above blocks closing M1.

16. Artifacts

Artifact	Path
Wrapper-stripping module	src/normalize.py
Normalize unit tests	tests/test_normalize.py
EDA notebook	notebooks/m1_eda.ipynb
Workstream A v1 local notebook	notebooks/m1_workstream_a_v1.ipynb
Kaggle inference notebook (with diagnostics)	notebooks/kaggle_inference_v1.ipynb
Baseline metrics notebook	notebooks/DL-23f2001849-milestone1.ipynb
Revised findings	reports/dataset_findings_revised.md
This report	reports/m1_report.md

GitHub: github.com/23f2001849/dl-genai-t22026-23f2001849 (main branch, milestone work committed inline).

Kaggle submissions: filter user [krishnaanalyst](#) at the competition submissions page.

W&B;: not used in M1 (Workstream A is deterministic). Activation begins in M2 with the first Workstream B training run.